

(width, height), resulting in a volume such as [16x16x12]

5. Fully-connected (FC) layer computes class scores, resulting in a volume of size [1x1x10], where each of the ten numbers correspond to a class score, such as among the ten categories of CIFAR-10. Each neuron in this layer is connected to all the neurons in the previous volume.

Thus, CNNs transform the original image layer by layer from the original pixel values to the final class scores. Note that some layers contain parameters, while others don't. In particular, Conv/FC layers perform transformations that are a function of not only activations in the input volume but also of neuron parameters (weights and biases). On the other hand, Relu/Pool layers implement a fixed function. Parameters in Conv/FC layers are trained with gradient descent so that class scores computed by the CNN are consistent with labels in the training set for each image.

Architecture for vehicle control

In end-to-end learning system for self-driving cars, weights of the network are trained to minimise the mean-squared error between the steering command output by the network and the command of either the human driver or the adjusted steering command for off-centre and rotated images. Fig. 2 shows the network architecture for vehicle control, which consists of nine layers, including a normalisation layer, five convolutional layers and three fully connected layers.

The input image is split into YUV planes and passed to the network. The network has about 27 million connections and 250,000 parameters. The first layer of the network performs image normalisation; the normaliser is hard-coded and not adjusted in the learning process. Performing normalisation in the network allows the normalisation scheme

to be altered with the network architecture, and accelerated via GPU processing.

The convolutional layers are designed to perform feature extraction, and are chosen empirically through a series of experiments that vary layer configurations. This model uses strided convolutions in the

Holistic neural network features

- Saliency
- Reduction, abstraction, induction, and abduction
- Generality
- Robustness
- Self-repair
- Novelty
- Scale-free
- Understanding

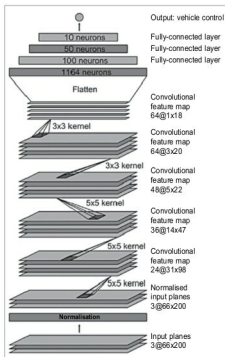


Fig. 2: CNN architecture for vehicle control

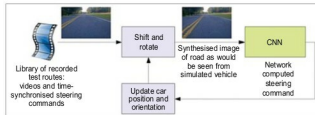


Fig. 3: Drive simulator

first three convolutional layers with a 2×2 stride and a 5×5 kernel, and a non-strided convolution with a 3×3 kernel size in the final two convolutional layers. The five convolutional layers are followed with three fully connected layers, leading to a final output control value which is the inverse-turning-radius. The fully connected layers are designed to function as a controller for steering, but by training the system end-to-end, it is not possible to make a clean break between which parts of the network function primarily as feature extractor, and which serve as controller.

Training and simulator

The first step in training a neural network is to select the frames for use. The collected data is labeled with road type, weather condition and the driver's activity (staying in a lane, switching lanes, turning and so forth). To train a CNN to do lane following, data is selected where the driver is staying in a lane, while the rest is discarded.

Then, the video is sampled at a rate of 10 frames per second because a higher sampling rate would include images that are highly similar, and thus not provide much additional useful information. To remove a bias towards driving straight, the training data includes a higher proportion of frames that represent road curves.

After selecting the final set of frames, the data is augmented by adding artificial shifts and rotations to teach the network how to recover from a poor position or orientation. The magnitude of these perturbations is chosen randomly from a normal distribution. The distribution has zero mean, and the standard deviation is twice the standard deviation that is measured with human drivers. Artificially augmenting the data does add undesirable artifacts as the magnitude increases.

Fig. 4 shows a screenshot of the simulator in interac-